The Suez Canal: Comprehensive Analytical Report (2015–2024)

Executive Summary

The Suez Canal is a vital maritime corridor supporting global trade across Asia, Europe, the Middle East, and North Africa. Between 2015 and 2024, the Canal handled **11,348 vessel transits**, generating **$24.91B in revenue** and **$5.66B in profit**, confirming its strong financial performance.

However, the analysis reveals significant operational vulnerabilities, particularly persistent congestion and delays, with **88% of transits delayed**. While container ships are the primary profit driver, revenue concentration, rising volatility, and underperforming vessel segments expose the Canal to long-term risk.

This report evaluates traffic patterns, vessel and cargo performance, delay behavior, regional contribution, and strategic risks, and provides actionable business recommendations to improve resilience and efficiency.

1. Background and Strategic Importance of the Suez Canal

The Suez Canal connects the Mediterranean Sea and the Red Sea, providing a direct maritime route between major global markets. Over the 2015–2024 period, the Canal averaged **~3.1 transits per day**, demonstrating sustained strategic importance.

Its role spans:

- · Containerized trade
- · Energy shipments (oil, gas)
- · Bulk commodities (grain, coal)

This diversity makes the Canal essential to global supply chains, while also amplifying the impact of disruptions.

2. Traffic Structure and Operational Performance

2.1 Vessel Traffic Composition

Traffic is dominated by Container Ships, followed by Tankers, Bulk Carriers, and LNG Carriers:

- · Container Ships: ~4.8K transits
- · Tankers: ~3.3K transits
- · Bulk Carriers: ~2.5K transits
- · LNG Carriers: ~0.7K transits

Container ships form the operational backbone of the Canal.

2.2 Delay Performance

Operational delays represent the most critical challenge:

- · Delayed transits: 10,010
- · On-time transits: 1,338
- · Delay rate: ~88%

Average delay by vessel type

- · LNG Carriers: 5.41 days
- · Tankers: 5.30 days
- · Bulk Carriers: 4.96 days
- · Container Ships: 4.63 days

Despite high volume, container ships remain the most efficient segment.

3. Financial Performance Analysis

3.1 Overall Financial Results (2015–2024)

- · Total Revenue: $24.91B
- · Total Cost: $19.24B
- · Total Profit: $5.66B
- · Profit Margin: ~22.7%

The Canal remains profitable, but profits are unevenly distributed.

3.2 Revenue Volatility

Recent trends indicate instability:

- · 2024 YoY revenue change: –34.9%
- · MoM decline: –52%

This reflects exposure to demand shifts, operational inefficiencies, and external shocks.

4. Vessel and Cargo Profitability

4.1 Profit by Vessel Type

- · Container Ships: $3.24B (57% of total profit)
- · Tankers: $1.64B
- · Bulk Carriers: $750M
- · LNG Carriers: $31M (<1% of profit)

Profit concentration is high, increasing systemic risk.

4.2 Cargo Revenue Contribution

Top cargo types by revenue:

- · Grain: $4.31B
- · Coal: $4.30B
- · Gas: $4.11B

Bulk commodities remain a major revenue pillar, with grain as the leading contributor.

5. Geographic Contribution

5.1 Transit Distribution by Region

- · Middle East: 2,361
- · North Africa: 2,329
- · Europe: 2,235
- · East Asia: 2,212
- · Asia: 2,211

5.2 Revenue Share by Region

- · Middle East: 21.1%
- · North Africa: 20.8%
- · Asia: 19.5%
- · East Asia: 19.4%
- · Europe: 19.3%

Revenue is geographically balanced, reducing dependency on a single region.

6. Analytical Questions and Findings

6.1 Why Is the Suez Canal a Critical Chokepoint?

- The canal handled approximately **11,348 vessel transits** during the analyzed period, highlighting its role as one of the busiest maritime corridors in the world.
- Alternative routes, particularly via the Cape of Good Hope, significantly increase voyage distance, transit time, fuel consumption, and insurance costs.
- Container vessels generate approximately **57% of total canal revenue**, making them the dominant contributor to profitability.

6.2 Cost Comparison with Alternative Routes

- Even when delays occur, the total cost impact is typically lower than the cumulative fuel, crew, and time costs of longer alternative routes.
- The continued high volume of transits during congestion periods suggests that operators prefer tolerating delays over incurring predictable and substantially higher rerouting costs.

6.3 Which Periods Show Highest Congestion?

- Data indicates no strong seasonal concentration of congestion, suggesting that high demand is structural rather than cyclical.
- Sustained high numbers of transiting vessels create a constant pressure on canal capacity, leaving limited room for operational flexibility.

6.4 Can Delays Be Predicted?

- Patterns in vessel type, traffic density, and historical delay distributions allow for moderate forecasting accuracy.

- Events such as groundings, geopolitical tensions, or extreme weather conditions introduce sudden disruptions that historical models cannot anticipate.

6.5 Main Risk Factors Contributing to Delays

- Traffic congestion
- Vessel-type constraints
- Operational inefficiencies
- External shocks (geopolitical, environmental)

6.6 Year-over-Year Trends

- Analysis shows no sustained reduction in average delay durations, indicating limited long-term improvement in congestion management.
- Recent years exhibit increased fluctuations in canal revenue, reflecting sensitivity to global trade disruptions and geopolitical events.
- While traffic levels remain strong, persistent delays and revenue instability suggest growing systemic risk.

7. Business Recommendations (Future Strategy)

- Shift focus from capacity expansion to efficiency optimization
- Protect and prioritize container ship operations
- Reassess LNG carrier pricing and routing strategy
- Implement predictive congestion analytics
- Introduce delay-based pricing and off-peak incentives
- Strengthen coordination with high-exposure regions
- Improve operational data transparency
- Align efficiency improvements with sustainability goals

8. Future Enhancements

Future analysis would benefit from:

- Real-time vessel tracking
- Weather and incident data
- Route-level freight cost information
- Port-level congestion metrics

- Fuel consumption and emissions data
- These would enable more precise delay forecasting and cost quantification.

9. Limitations

- No direct freight rate or fuel cost data
- Limited temporal granularity (no daily/weekly delays)
- No explicit weather or incident variables
- Port-specific impacts inferred, not directly measured
- Despite these constraints, the dataset provides strong strategic insights.

10. Conclusion

Between 2015 and 2024, the Suez Canal demonstrated sustained financial strength despite persistent operational constraints. High-value vessel segments—particularly container shipping and bulk commodities such as grain, which generated approximately **$4.31 billion**—remain the primary drivers of revenue. However, this strong financial performance masks underlying structural vulnerabilities, including chronic transit delays, heavy revenue concentration within specific vessel categories, and increasing revenue volatility.

These dynamics highlight a growing risk profile. Continued reliance on traffic volume alone is unlikely to ensure long-term stability, especially in the face of geopolitical disruptions, environmental shocks, and evolving global trade patterns. Instead, the canal's long-term resilience will depend on targeted improvements in operational reliability, transit efficiency, and real-time data transparency. Strengthening these areas is essential not only to reduce delays but also to preserve the canal's strategic relevance in an increasingly competitive and risk-sensitive global shipping environment.